

Spatial-Temporal Masked Autoencoder for Multi-Device Wearable Human Activity Recognition

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The widespread adoption of wearable devices has led to a surge in the development of multi-device wearable human activity recognition (WHAR) systems. Nevertheless, the performance of traditional supervised learning-based methods to WHAR is limited by the challenge of collecting ample annotated wearable data. To overcome this limitation, self-supervised learning (SSL) has emerged as a promising solution by first training a competent feature extractor on a substantial quantity of unlabeled data, followed by refining a minimal classifier with a small amount of labeled data. Despite the promise of SSL in WHAR, the majority of studies have not considered missing device scenarios in multi-device WHAR. To bridge this gap, we propose a multi-device SSL WHAR method termed Spatial-Temporal Masked Autoencoder (STMAE). STMAE captures discriminative activity representations by utilizing the asymmetrical encoder-decoder structure and two-stage spatial-temporal masking strategy, which can exploit the spatial-temporal correlations in multi-device data to improve the performance of SSL WHAR, especially on missing device scenarios. Experiments on four real-world datasets demonstrate the efficacy of STMAE in various practical scenarios.

CCS Concepts: • **Human-centered computing** → **Mobile computing**; • **Computing methodologies** → **Neural networks**.

Additional Key Words and Phrases: human activity recognition, wearable sensors, self-supervised learning, annotation scarcity

ACM Reference Format:

Shenghuan Miao, Ling Chen, and Rong Hu. 2023. Spatial-Temporal Masked Autoencoder for Multi-Device Wearable Human Activity Recognition. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 7, 4, Article 172 (December 2023), 25 pages. <https://doi.org/10.1145/3631415>

1 INTRODUCTION

The popularity of wearable devices has promoted the booming development of wearable human activity recognition (WHAR), and its application areas cover education [20, 21], sports [36, 38], physical health [7, 12, 67, 80, 81], etc. WHAR can be divided into single-device WHAR and multi-device WHAR based on the number of devices the user is wearing. Typically, each device records data from several modalities. For example, a common IMU device records the values of acceleration, angular velocity, and magnetic field [59]. Recently, the development of Virtual and Augmented Reality (VR/AR) kicks off a new storm about multi-device WHAR, which is a more sophisticated and practical scenario [2, 86]. By capturing activity information from multiple body parts at the

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2474-9567/2023/12-ART172 \$15.00

<https://doi.org/10.1145/3631415>

same time, multiple devices are certainly beneficial for improving the recognition performance, especially for multi-joint activities, e.g., distinguishing the activity between “running” and “cycling” [1, 9, 10, 43, 48, 52, 84].

Supervised learning is widely adopted in WHAR, i.e., a model is first trained on a labeled dataset and then applied to real-world data [53]. The performance of supervised learning methods is particularly dependent on large-scale labeled datasets, which require significant human efforts and financial costs [62]. Unlike image or text data that can be annotated through crowdsourcing or other means after data collection, wearable data require annotation at the time of data collection. This renders activity labeling difficult, reducing the number of available labels [8, 23, 30]. Self-supervised learning (SSL) provides a viable solution by first pre-training a feature extractor on a large amount of unlabeled data and then optimizing a small classifier with limited labeled data [13]. The differences between diverse SSL WHAR methods are mainly in the design of pre-training tasks. Existing pre-training tasks for SSL WHAR can be mainly divided into three categories: hand-crafted auxiliary tasks [61, 63, 76], contrastive learning tasks [14, 26, 34, 40, 51, 54, 58, 64], and masked reconstruction tasks [25, 56, 71], with contrastive learning tasks occupying the majority. Hand-crafted auxiliary tasks design additional objectives based on heuristics to provide the model with supervisory signals [61, 63, 76]. Contrastive learning tasks aim to maximize agreement of an instance and its positive samples while minimizing agreement between the instance and its negative samples in feature space [14, 26, 34, 40, 51, 54, 58, 64]. Masked reconstruction tasks leverage encoders to extract valid features from partially masked input and decoders to recover the complete input by learning the correlation patterns within the data [25, 56, 71].

While off-the-shelf SSL WHAR methods have shown promising results, the majority of them have been conducted in single-device scenarios. Although we can directly concatenate the raw data from multiple devices as the input to existing single-device SSL WHAR methods, doing so ignores the spatial information between multiple devices. Many studies have shown that capturing such spatial information contributes to the extraction of effective features [1, 34, 43, 48, 84]. Most recently, there are two preliminary contrastive learning based studies [14, 51] on multi-device scenarios. Contrastive learning is vulnerable to biased input instances, as it strongly relies on data augmentation methods to generate positive and negative samples. It is noteworthy that there exists inter-activity similarity and intra-activity variability in WHAR. For example, the wearable readings of activities “climbing up” and “running” are relatively similar, with similar value ranges and periodical patterns. Instead, two independent readings from the same activity “climbing up” may have different value ranges and periodical patterns. The inter-activity similarity and intra-activity variability imply instances are biased easily in WHAR, which exacerbate the difficulty of constructing positive and negative samples in contrastive learning.

Most importantly, existing SSL WHAR methods do not consider missing device scenarios in multi-device WHAR. For example, during both pre-training and supervised learning phases, we can train the WHAR model with complete multi-device data, containing the devices of smart glasses (head), smartphone (wrist), smart jacket (back, upper arm, lower arm), and smart shoes (feet). In different scenarios, the user may wear different devices. The user may take off his/her smart shoes and smart glasses when at home, and may take off his/her smart jacket when the weather is hot. This situation is realistic, as it is impossible for every user to wear all devices at all time in daily life. In addition, due to the changing characteristics of the devices (e.g., lack of power and loss of signal), only part of devices may be available [19, 33, 39, 42, 45]. As a result, the available devices are an arbitrary subset of the total devices, which is unpredictable and would severely impact the recognition performance of existing methods.

To address the above issues, we propose a multi-device SSL WHAR framework termed Spatial-Temporal Masked Autoencoder (STMAE), which tackles missing device scenarios through a further exploration in the design of masking strategy and encoder-decoder structure [25, 28, 56, 71]. In a nutshell, our contributions are summarized as follows:

- We introduce a two-stage spatial-temporal masking strategy. First, raw data from certain devices are completely discarded in spatial masking to simulate missing device scenarios. Then, an extremely high masking probability is adopted in temporal masking to facilitate the encoders in learning temporal patterns. Such a masking strategy design can handle missing device scenarios effectively.
- We introduce an asymmetrical encoder-decoder structure comprising relatively large encoders, i.e., Spatial Encoder (SE) and Temporal Encoder (TE), and a relatively small decoder, i.e., Reconstruction Decoder (RD). Such an encoder-decoder structure design can impose a greater demand on the encoders for learning spatial-temporal correlations that benefits downstream tasks.
- Our evaluation on four public datasets demonstrates that STMAE outperforms the state-of-the-art (SOTA) baselines. In addition, we conduct in-depth analyses to understand the strengths and limitations of STMAE.

2 RELATED WORK

In this section, we provide a review of two sets of relevant prior research within the larger field of WHAR. Specifically, we begin by discussing the general background of multi-device WHAR, and then we delve into the topic of SSL for WHAR. This review allows us to situate our work within existing body of research and to demonstrate the significance of our study.

2.1 Multi-Device WHAR

Multi-device WHAR is widely used in various scenarios to understand the complex and changing human activities. Di Lascio et al. [16] propose a method for tracking breaks and work periods in academia using multiple devices from smartwatches, laptops, and smartphones. NeckSense [82] considers building a multi-device necklace for detecting eating activities and episodes in free-living conditions. DeepIT [22] introduces a multi-stage device fusion pipeline to realize inertial motion tracking by synthesizing the IMU measurements from a smartphone and an associated earbud. These examples demonstrate the versatility and utility of multi-device WHAR in different contexts.

2.1.1 Supervised Learning for Multi-Device WHAR. Deep learning has paved the way for many new multi-device WHAR methods, with a focus on improving recognition performance. DeepConvLSTM [50] first concatenates the multi-device data from each device together. Then it mines the temporal information successively by convolutional neural network (CNN) and recurrent neural network (RNN) while ignores the multi-device correlations learning. DeepSense [73] uses CNN for multi-device information extraction and RNN for temporal information extraction. In addition to the CNN and RNN, multi-device WHAR continues to evolve with the advancement of deep learning technologies, e.g., attention mechanism [66] and Graph Convolutional Network (GCN). SADeepSense [74] extends DeepSense [73] by introducing the self-attention mechanism [66] to learn the correlations among different devices. The attention mechanism [66] is also used by AttnSense [46], GlobalFusion [43], Attend-Discriminate [1], and IF-ConvTransformer [84] to explore multi-device correlations learning. DynamicWHAR [48] designs a GCN-based dynamic inter-sensor correlations learning framework for multi-device WHAR. While the leaderboard may be updated with the latest methods, the value of this progress is limited due to the use of supervised learning methods that rely on labeled data for training. As we have demonstrated in Section 1, there is a scarcity of labeled data in the field of WHAR. Therefore, it is crucial to investigate SSL as a potential solution.

2.1.2 Missing Device Scenarios in Multi-Device WHAR. Multi-device WHAR offers various benefits, but it also introduces the problem of missing devices. This is an inevitable and widespread issue. Previous research has explored methods for dealing with missing devices in WHAR. Zappi et al. [78] investigate the recognition performance with varying number of devices by a meta-classifier that fuses the information of simple classifiers operating on individual devices. Fekade et al. [19] propose a two-step method, where it first uses K-means

clustering to split the devices into different groups, and then performs probabilistic matrix factorization within these groups to recover missing data. Imani et al. [33] present a nested Gaussian process for missing data imputation. Zhou et al. [85] propose an iterative imputing network for recovering missing device data, which captures latent complex temporal dynamics. While the above methods are designed for the specific missing device scenario independently, some methods have been developed that can handle arbitrary missing device scenarios. ArmTroi [45] uses an attention-based design to automatically adjust the weights of valid and missing devices. Vaizman et al. [65] randomly mask some devices as "missing" during training, and amplify the features of the available devices to maintain their total contribution. Jaques et al. [35] propose the use of a denoising autoencoder for handling missing device scenarios. Saeed et al. [60] and Yoon et al. [75] use the adversarial autoencoder and the generative adversarial network, respectively, for imputing missing devices. Liu et al. [42] and Jiang et al. [37] explore the use of graph neural networks for this purpose. AALH [39] proposes an adversarial domain adaptation framework with two augmentation strategies for solving the problem of missing devices. It is difficult to directly compare the above methods, as the assumed missing device scenarios vary widely across different papers. The goal of our work is not to propose a new method for dealing with missing devices, but to consider this scenario in the design of the pre-training task in SSL.

2.2 SSL for WHAR

SSL has sparked a surge of interest in many fields because of its ability to learn directly from unlabeled data. In particular, in the field of natural language processing, large pre-trained language models, e.g., Bidirectional Encoder Representation from Transformers (BERT) [15] and Generative Pre-Training (GPT) [55], have achieved impressive results in many tasks, e.g., text generation, language understanding, text classification, and question answering. A variety of SSL methods have emerged for learning general representations of time series that can be applied to downstream tasks, e.g., classification, forecasting, and anomaly detection [32, 47, 69, 72, 77, 79, 83]. Recently, there has been notable advancement in learning general representations of images [3, 28], videos [18], and audio [31] through mask reconstruction-based SSL methods. While significant progress has been made in existing studies, it is important to pursue SSL designs specifically tailored to the intrinsic characteristics of WHAR dataset.

SSL has also been widely explored in WHAR. Accelerometer features are learned by training a feature extractor to recognize transformations applied to the raw input signals, e.g., "noised", "scaled", "rotated", "negated", "horizontally flipped", "permuted", "time-warped", and "channel-shuffled" [61]. Yuan et al. [76] extend this idea by using multi-task learning with the data transformations of "arrow of time", "permutation breaks", and "time warping". Tang et al. [64] explore the adoption and adaptation of SimCLR [11], a representative contrastive learning technique for visual representations, to WHAR, using time-series transformation methods in place of image augmentation operators. After [64], subsequent works in this area have largely focused on contrastive learning methods. SimCLR [11] is also used in [40] with a transformer-based encoder as the feature extractor. Haresamudram et al. [26] adopt the Contrastive Predictive Coding (CPC) [49] framework, which involves predicting future timesteps of device data from a context vector derived from past data by autoregressive modeling. Haresamudram et al. [26] demonstrate that considering the temporality of device data contributes to effective learning of the feature extractor. Qian et al. [54] conduct a thorough study of contrastive learning for WHAR through both algorithmic-level and task-level investigations.

Our work is most closely related to the pre-training tasks of masked reconstruction [25, 56, 71]. Haresamudram et al. [25] use a variant of BERT [15] that replaces 10% of the wearable data at random time steps with zeros, and then uses mean squared error loss to reconstruct the masked data. Xu et al. [71] propose LIMU-BERT, which effectively captures temporal relations and feature distributions from unlabeled IMU data through effective normalization, span masking, and a lightweight design. Rahimi et al. [56] use cross-dimensional motion prediction

as the pre-training task. Specifically, Rahimi et al. [56] leverage past and present accelerometer data of the x and y dimensions, as well as past accelerometer data of the z dimension to predict the present values of the z dimension. Haresamudram et al. [27] systematically evaluate SSL methods for WHAR on three criteria: "Robustness to differing source and target conditions", "Influence of dataset characteristics", and "Feature space characteristics". While previous research on SSL for WHAR has demonstrated promising results, these studies have largely focused on single-device scenarios and have not considered the challenges and opportunities presented by multi-device data.

Recently, several works [14, 34, 51, 58] have emerged that consider multi-device scenarios. Jain et al. [34] believe that unlabeled wearable datasets simultaneously captured by multiple devices can be viewed as natural transformations of each other. The authors propose ColloSSL, which demonstrates that learning the correlations across multiple devices benefits contrastive learning. ColloSSL incorporates three innovations: "Device Selection", "Contrastive Sampling", and "Multi-view Contrastive Loss". Rey et al. [58] propose to transfer useful information from the source devices to the latent representation of the target device data by constructing the contrastive pairs from the source devices and the target device. While ColloSSL [34] and [58] use multi-device data to empower the pre-training of the feature extractor, both of them are designed for single-device scenarios in downstream tasks. Cosmo [51] integrates fusion-based contrastive learning and quality-guided attention mechanisms to effectively extract consistent and complementary information from different modalities. COCOA [14] is a contrastive learning framework for learning general-purpose representations from multimodal time-series data, using a cross-modal contrastive objective function that is more computationally efficient than other multimodal contrastive learning methods. While these contrastive learning-based methods have made promising progress in multi-device scenarios, none of them consider the missing device scenarios.

3 METHOD

In this section, we first define the multi-device SSL WHAR problem formally. Then we describe the overall training and deployment workflow of STMAE. Finally, we elaborate the asymmetrical encoder-decoder structure and masking strategy in STMAE.

3.1 Problem Definition

Let the unlabeled dataset be $\mathcal{D}^u = \{X_i \mid i = 1, \dots, L^u\}$ and labeled dataset be $\mathcal{D}^l = \{(X_i^l, y_i) \mid i = 1, \dots, L^l\}$ with $L^u \gg L^l$, where L^u and L^l denote the number of unlabeled samples and labeled samples, respectively. Unlabeled sample $X_i = (X_{i,1}, X_{i,2}, \dots, X_{i,N}) \in \mathbb{R}^{N \times T \times C}$, where N is the number of devices, T is the window size, C is the number of raw readings from the device. For the device j of X_i , $X_{i,j} = (x_{i,j,1}, x_{i,j,2}, \dots, x_{i,j,T}) \in \mathbb{R}^{T \times C}$. Labeled sample X_i^l has the same data dimensions as X_i . y_i is the corresponding activity label for X_i^l . The problem is how to use $\mathcal{D}^u$ and $\mathcal{D}^l$ to train a WHAR model that has strong discriminative ability for unseen samples. We first use $\mathcal{D}^u$ to pre-train the feature extractor $\mathcal{F}$. The optimized parameters Θ for the feature extractor $\mathcal{F}(\cdot, \Theta)$ are fixed after pre-training. The classifier $\mathcal{G}$ learns its optimized parameters Φ by using $\mathcal{D}^l$ in a supervised manner. Finally, the feature extractor $\mathcal{F}(\cdot, \Theta)$ can extract the generalizable representations and the classifier $\mathcal{G}(\cdot, \Phi)$ can distinguish the activities for unseen samples.

3.2 Overview

The overall architecture of STMAE is shown in Fig. 1. STMAE takes the advantage of the asymmetrical encoder-decoder structure with two-stage spatial-temporal masking strategy to exploit the spatial-temporal correlations in multi-device data. The learned spatial-temporal correlations can help to extract versatile activity representations even when only part of devices are available, which is beneficial for downstream tasks. The training and deployment of STMAE can be divided into the following three phases:

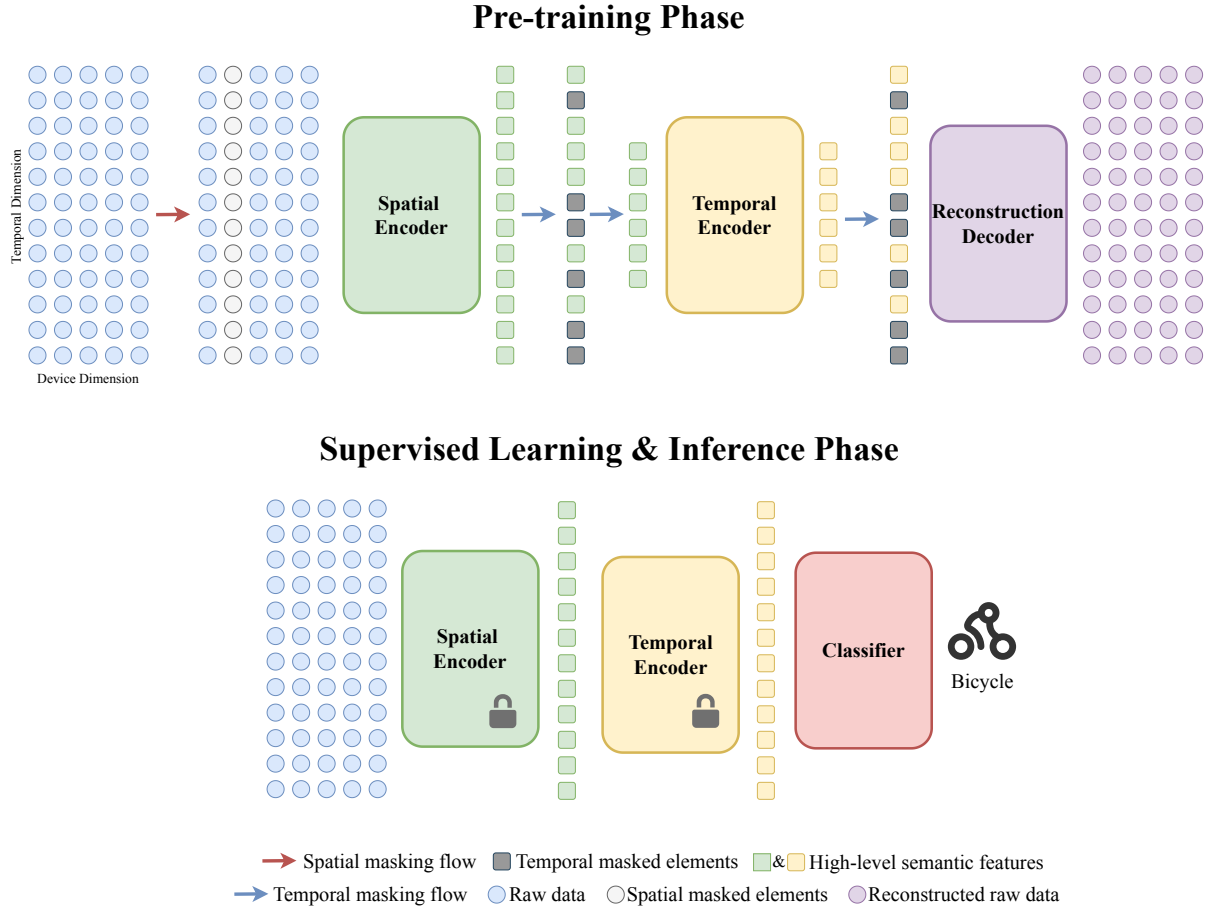


Fig. 1. STMAE Overview. STMAE takes the advantage of the asymmetrical encoder-decoder structure with two-stage spatial-temporal masking strategy to exploit the spatial-temporal correlations in multi-device data. Please refer to Section 3 for a detailed explanation. (Best viewed in color.)

Pre-training phase. Spatial Encoder (SE) and Temporal Encoder (TE) encode the spatial information and temporal information for the unlabeled sample X_i , respectively. Spatial masking and temporal masking are used before SE and TE to mask part of the input data, respectively. Then, Reconstruction Decoder (RD) is designed to recover the masked data.

Supervised learning phase. The labeled sample X_i^1 passes through SE and TE sequentially without any masking strategies. Then, the extracted spatial-temporal features will be fed into the downstream classifier. The parameters of SE and TE are frozen, only the parameters of the classifier are optimized. After supervised learning, the classifier can use the learned spatial-temporal features to predict the corresponding activity.

Inference phase. No parameters of STMAE need to be updated anymore. STMAE recognizes human activities by passing the real-world wearable data through SE, TE, and the classifier sequentially.

3.3 Spatial Masking and Encoding

In the context of multi-device data, comprehending the spatial correlations between data from various wearable devices is essential for determining the current activity. This is because the spatial correlations differ depending on the activity being performed. These spatial correlations can also be utilized to infer data from one device based on the data from other devices, making them a powerful means of reducing data redundancy. By taking advantage of these spatial correlations, we are able to achieve effective feature extraction even in cases where data from certain devices are unavailable. To accomplish this, we aim to implement practical spatial masking and encoding techniques to capture these valuable spatial correlations.

The design of our spatial masking is straightforward and simple. We randomly select N_s out of N devices in each input sample such that all data from the selected devices are masked. This simulates the real-world scenario where some devices are not available during inference phase. To indicate which elements should be masked in X_i , we define the spatial masked matrix M_s that has the same shape as X_i . M_s can be expressed as follows,

$$M_s^{i,j,k} = \begin{cases} 0 & i \in \text{RANDOM_INDEX}(N_s, N) \\ 1 & \text{otherwise} \end{cases} \quad (1)$$

where $\text{RANDOM_INDEX}(N_s, N)$ randomly samples N_s integers from the range of $[0, N)$ without replacement. The spatial masking number N_s is set to 1. The input sample after spatial masking is obtained by taking the dot product of M_s and X_i , which will be fed to SE to perform the extraction and fusion of spatial features.

The goal of SE is to convert raw wearable data into high-level semantic features that are suitable for neural network processing, while also fusing features from individual devices at every time point. The initial spatial feature extraction and fusion of raw wearable data will bring convenience to the subsequent capturing of temporal correlations. When designing SE, it is necessary to keep in mind the inherent properties of wearable data. To ensure wearing comfort, the number of devices that people carried is generally low, typically less than ten [6, 57, 59, 62]. Each device has a relatively small number of raw features. For example, an IMU device typically contains raw features from a 3-axis accelerometer, a 3-axis gyroscope, and a 3-axis magnetometer. However, wearable data contain a relatively large number of temporal features and rich temporal context information. For example, a 2-seconds input sample at a common sampling rate of 50 Hz contains 100 time points in the time dimension. It means that the features in the time dimension are an order of magnitude larger than those in the spatial dimension. Based on the above characteristics of wearable data, we choose to use the CNN to perform the initial fusion of multi-device data to express richer spatial information. Specifically, we use a layer of CNN that contains D_{se} 2D convolution kernels of size $(N, 5)$ to aggregate the features of each device at the same time point,

$$X_i^s = \text{ENCODER}_{se}(M_s \odot X_i) \quad (2)$$

where $X_i^s \in R^{T \times D_{se}}$ is the output of ENCODER_{se} . X_i^s reserves the time dimension while the raw features from individual devices are fused and extended from C to D_{se} . The mechanism of spatial masking and encoding allows the raw wearable data to be fused spatially even in the absence of some device data, contributing to the effective feature extraction at each time point.

3.4 Temporal Masking and Encoding

Spatial masking primarily simulates missing device scenarios, while the role of temporal masking is to prepare for learning temporal correlations by TE. P_t of the time points are masked in temporal masking, where P_t is the temporal masking probability. The number of time points to be masked is the rounded-down product of P_t and T .

We define the temporal masked matrix $\mathbf{M}_t$ that has the same shape as $\mathbf{X}_i^s$, which can be expressed as follows,

$$\mathbf{M}_t^{i,j} = \begin{cases} 0 & i \in \text{RANDOM_INDEX}(\lfloor P_t \times T \rfloor, T) \\ 1 & \text{otherwise} \end{cases} \quad (3)$$

where $\text{RANDOM_INDEX}(\lfloor P_t \times T \rfloor, T)$ randomly samples $\lfloor P_t \times T \rfloor$ integers from the range of $[0, T)$ without replacement. It is important to note that due to the high sampling rate of wearable devices (typically 50 Hz~200 Hz), the raw readings at two adjacent time points have a strong similarity. We need to prevent TE and RD using the information from the neighboring available time points directly as the information of masked time points that needs to be recovered. Fortunately, the temporal information we need to mask now is not the raw readings, but the high-level semantic features after spatial fusion. In addition, we set the temporal masking probability P_t to a higher value (e.g., 50%) to increase the challenge of reconstruction. Furthermore, we skip masked elements and only apply temporal encoding to unmasked elements. It also helps reduce computational costs during pre-training. The input to TE can be derived as follows,

$$\mathbf{X}_i^{\text{ms}} = \text{NONE_ZERO}(\mathbf{M}_t \odot \mathbf{X}_i^s) \quad (4)$$

where $\text{NONE_ZERO}(\mathbf{M}_t \odot \mathbf{X}_i^s)$ ensures that $\mathbf{X}_i^{\text{ms}}$ only contains the non-zero rows of $\mathbf{X}_i^s$.

Since different activities exhibit different periodic patterns, amplitude ranges, etc., in the time series, capturing temporal correlations is crucial to obtain features that can fully characterize the current activity. TE aims to mine as many discriminative temporal features in $\mathbf{X}_i^{\text{ms}}$ as possible. Specifically, we use the structure of Transformer encoder [66] to fully exploit temporal correlations. In order to preserve the temporal order information of $\mathbf{X}_i^{\text{ms}}$, we first add the standard 1D sine-cosine position embeddings [66] $\mathbf{E}_{\text{pos}}$ to $\mathbf{X}_i^{\text{ms}}$,

$$\mathbf{Z}_0 = \mathbf{X}_i^{\text{ms}} + \mathbf{E}_{\text{pos}} \quad (5)$$

Then, $\mathbf{Z}_0$ goes through L_{te} Transformer encoders in sequence. For the Transformer encoder ℓ , the layer normalization (LN) version of $\mathbf{Z}_{\ell-1}$ goes through the multi-head self-attention (MSA) block and multilayer perceptron (MLP) block in order. In addition, residual connection is applied after every MSA block or MLP block. $\mathbf{Z}_\ell$ is the output of the Transformer encoder ℓ and the input of the Transformer encoder $\ell + 1$, which can be calculated as follows,

$$\mathbf{Z}'_\ell = \text{MSA}(\text{LN}(\mathbf{Z}_{\ell-1})) + \mathbf{Z}_{\ell-1}, \quad \ell = 1 \dots L_{\text{te}} \quad (6)$$

$$\mathbf{Z}_\ell = \text{MLP}(\text{LN}(\mathbf{Z}'_\ell)) + \mathbf{Z}'_\ell, \quad \ell = 1 \dots L_{\text{te}} \quad (7)$$

where $\text{MSA}(\cdot)$, $\text{MLP}(\cdot)$, and $\text{LN}(\cdot)$ is the MSA, MLP, and LN block, respectively. All the MSA blocks consist of H_{te} attention heads with the hidden dimensions of D_{te} . All the MLP blocks contain two fully-connected (FC) layers with the hidden dimensions of D_{te} . We treat $\mathbf{Z}_{L_{\text{te}}}$, the output of the Transformer encoder L_{te} , as the output of TE.

Unlike in spatial masking where masking part of input are directly set as zeros, in temporal masking we set the value of masking part as the mask token $[M]$, which is a learnable variable. The reason is a large number of zeros in time dimension will affect the subsequent temporal feature extraction for RD. By setting the masking component to a learnable value, it can force the model to learn how to best fill in the masked areas during pre-training phase. The output after temporal masking and encoding can be expressed as follows,

$$\mathbf{X}_i^t = \text{UNSHUFFLE}(\mathbf{Z}_{L_{\text{te}}}, \mathbf{M}_t) \quad (8)$$

where $\text{UNSHUFFLE}(\mathbf{Z}_{L_{\text{te}}}, \mathbf{M}_t)$ maps each row vector in $\mathbf{Z}_{L_{\text{te}}}$ back to the corresponding time point indices in $\mathbf{M}_t$ that are not masked, with $[M]$ being filled in the masked time point indices.

3.5 Reconstruction Decoder

After sufficient spatial-temporal information interactions with other time points, X_i^t is characterized more globally at each time point. RD aims to reconstruct the masked raw data using X_i^t , which can be formulated as follows,

$$X_i^{\text{rd}} = \text{DECODER}_{\text{rd}}(X_i^t) \quad (9)$$

If RD can recover the masked wearable values well, it means that the spatial and temporal encoders capture the spatial-temporal correlations well. Since the reconstructed output are still time series, we use L_{rd} Transformer encoders for RD. Each Transformer encoder contains H_{rd} attention heads with the hidden dimensions of D_{rd} . Note that the Transformer encoder we use in RD has a smaller number of parameters than in TE, leading to a asymmetric encoder-decoder structure of STMAE, for two reasons. On the one hand, we expect X_i^t to contain rich spatial-temporal information so that it can satisfy both the reconstruction task and other downstream tasks using a simple neural network structure. On the other hand, we expect the training cost in supervised learning phase to be mitigated. Lightweight downstream neural network structure is meaningful, as supervised learning phase is usually done on edge but not on cloud.

3.6 Loss Function

The loss function in pre-training phase is formulated as follows,

$$\mathcal{L}_{\text{pre-train}} = \frac{1}{L^u} \sum_i^{L^u} \text{MSE}(X_i, X_i^{\text{rd}}) \quad (10)$$

where $\text{MSE}(X_i, X_i^{\text{rd}})$ computes the mean square error of X_i and X_i^{rd} only on the masked part. The loss function in supervised learning phase is cross-entropy loss,

$$\mathcal{L}_{\text{fine-tune}} = \frac{1}{L^l} \sum_i^{L^l} \text{CROSS_ENTROPY}(y_i, \hat{y}_i) \quad (11)$$

where y_i and $\hat{y}_i$ is the true activity label and predicted activity label for X_i^l , respectively.

4 EXPERIMENTAL PROTOCOLS

In this section, we introduce the experimental protocols used in our study, including the datasets utilized, the baselines employed, and the specifics of experimental setup.

4.1 Datasets

We select four widely used public multi-device datasets (Opportunity [59], RealWorld [62], Realdisp [6], and PAMAP [57]) for evaluation. These datasets have distinct characteristics in the number of subjects, the number of devices, the number and type of activities, and the sampling frequency, which opens the door to a comprehensive and unbiased evaluation of our method.

Opportunity [59]: Opportunity dataset contains data collected from 4 participants using 7 IMUs and 12 3D acceleration sensors. We use the data collected at 30 Hz from 5 IMUs that deployed using a jacket, which placed on right up arm, right low arm, left up arm, left low arm, and back. The subjects performed activities of daily living in a breakfast scenario. The activities were annotated on different levels from coarse to fine. We target to detect 17 mid-level gesture classes for our evaluation.

RealWorld [62]: In RealWorld dataset, 8 physical activities of 15 users were collected using 7 mobile devices (six smartphones and a smartwatch) placed on head, chest, upper arm, waist, thigh, shin, and forearm, respectively.

Table 1. Statistical Summary of Datasets.

Dataset	# Devices	# Activities	# Subjects	Frequency	Window Size	Sample Size
Opportunity [59]	5	17	4	30 Hz	60	8,165
RealWorld [62]	7	8	13	50 Hz	150	39,281
Realdisp [6]	9	33	10	50 Hz	150	6,280
PAMAP [57]	3	12	8	100 Hz	300	12,902

We use 13 out of these 15 users for our experiments, as the activity labels of the remaining two users are incomplete. The data were recorded at 50 Hz.

Realdisp [6]: Realdisp dataset contains data of 33 different fitness activities, performed by 17 participants with 9 IMUs on left/right calf, left/right thigh, left/right lower arm, left/right upper arm, and back. Data from 10 users with sampling rate of 50 Hz are utilized in the evaluation due to the incomplete activity labels of remaining users. To study the impact of sensor placement, participants performed three data collection sessions: ideal-placement, self-placement, and mutual-placement. Only data from the ideal-placement session are used in our experiments, as the sensor placement problem goes beyond the scope of this paper.

PAMAP [57]: PAMAP dataset includes data from three IMUs and a heart rate monitor collected from 9 individuals during the performance of 12 realistic activities. The IMUs were placed on the dominant wrist, chest, and dominant ankle with 100 Hz sampling rate. Our experiments are performed using the data of 8 users, as the data for the remaining user are seriously insufficient.

Following [34], the raw data are segmented into 2-seconds intervals for Opportunity dataset and 3-seconds intervals for RealWorld dataset. For Realdisp and PAMAP datasets, the lengths of segmentation are also set as 3-seconds for fair comparison. Hence, the window size T is 60/150/150/300 for Opportunity, RealWorld, Realdisp, and PAMAP, respectively. After sliding window segmentation with 50% overlap, the sample size is 8,165/39,281/6,280/12,902 for Opportunity, RealWorld, Realdisp, and PAMAP, respectively. To all the datasets, we use the readings from 3-axis accelerometer, 3-axis gyroscope, and 3-axis magnetometer for each device. Hence, both X_i and X_i^1 have the shape of $\mathbb{R}^{5 \times 60 \times 9}$, $\mathbb{R}^{7 \times 150 \times 9}$, $\mathbb{R}^{9 \times 150 \times 9}$, $\mathbb{R}^{3 \times 300 \times 9}$ for Opportunity, RealWorld, Realdisp, and PAMAP, respectively. The statistical summary of datasets are given in Table 1. The detailed activity types and their distributions in each dataset are provided in Appendix A. Due to the specificity of the reconstruction task in our work, we do not use any data normalization methods.

4.2 Baselines

We compare the performance of STMAE with seven representative baselines, which can be divided into the following three categories:

Simplified STMAE: To demonstrate the necessity and effectiveness of SSL, we devise two simplified versions of STMAE while maintaining its network structure.

- **STMAE-Sup:** STMAE-Sup is the supervised version of STMAE. The parameters of TE, SE, and classifier in STMAE are jointly optimized using labeled dataset $\mathcal{D}^l$, while the unlabeled dataset $\mathcal{D}^u$ will be ignored.
- **STMAE-Rand:** In STMAE-Rand, the parameters of TE and SE are given randomly instead of pre-training. The parameters of classifier are optimized with labeled dataset $\mathcal{D}^l$.

Masked Reconstruction for SSL: Since the pre-training task of STMAE belongs to the category of masked reconstruction, the comparison with the same kind methods can demonstrate our innovation on masking strategies and encoder-decoder network.

- **Masked Reconstruction [25]:** Masked Reconstruction randomly chooses 10% of the time steps and set their values to zeros. Data after masking are fed into Transformer encoders for the reconstruction.
- **LIMU-BERT [71]:** LIMU-BERT adapts BERT to best work with IMU data by introducing the new normalization methods, the span masking mechanism, and the lightweight network.

Contrastive Learning for SSL: As the most widely used and powerful method currently available, comparing with the pre-training tasks of contrastive learning will help us to identify the upsides and downsides of our work. In addition, COCOA [14] and Cosmo [51] are the few methods currently available that can directly work for multi-device WHAR.

- **COCOA [14]:** COCOA aims to maximize the agreement between modalities at the same time step and minimize the agreement between randomly sampled and temporally distant samples.
- **Cosmo [51]:** Cosmo integrates novel fusion-based contrastive learning and quality-guided attention mechanisms to effectively extract both consistent and complementary information across different modalities.
- **CPC [26]:** CPC involves predicting future time steps of device data from a context vector derived from past data by autoregressive modeling. The InfoNCE loss [49] between predictions and ground truth is utilized to optimize the weights of feature extractor.

4.3 Experimental Settings

In the typical deployment of a WHAR model [51], the pre-training phase is performed in the cloud to leverage the cloud computing power and large-scale wearable datasets. The supervised learning phase is performed in the local devices (e.g., smartphones) to personalize the WHAR model while ensuring user privacy. In the supervised learning phase, users are guided to perform various activities to complete the data annotation. This guidance process is usually short for better user experience, so the amount of available labeled data are very limited. Leave-one-group-out cross-validation approach is used for performance evaluation follows the previous works [26, 34]. The number of groups are set as 4, 5, 5, and 4 for Opportunity, RealWorld, Realdisp, and PAMAP2, respectively. No data augmentation methods are used during any training phases. The labeling rate is 1% in the training set for Realdisp dataset, i.e., only 1% of the data in training set are labeled while 99% are unlabeled. Since the sample sizes in the remaining datasets are much smaller, we set the labeling rate to 10% for other datasets. The impact of labeling rate is analyzed in Section 5.2. The unlabeled data and labeled data from training set are used in pre-training phase and supervised learning phase, respectively. The test set is used in inference phase for the final performance evaluation. We adopt macro F1-score (unweighted mean of F1 scores over all classes) as the performance metric following [14, 34]. The final reported result is the average macro F1-score across five independent runs by varying seeds. For the implementation details, please refer to Appendix B.

5 EVALUATION AND DISCUSSION

In this section, we delve deeper into the essence of STMAE through a comprehensive quantitative and qualitative analysis of its real-world deployment and application scenarios.

5.1 Comparison with Baselines

We evaluate the performance of different methods on four datasets with limited labeled data following the experimental protocols in Section 4. The results are presented in Table 2. STMAE achieves the best performance on all datasets compared to other SSL baselines, which demonstrates the effectiveness of spatial-temporal masking and encoding. STMAE outperforms the best SSL baselines by 5.41%, 2.81%, 3.95%, and 5.00% on Opportunity, RealWorld, Realdisp, and PAMAP, respectively. On three out of four datasets, STMAE even achieves higher performance than the supervised method STMAE-Sup. In the remaining one dataset, the performance of STMAE is comparable to STMAE-Sup (with -0.79% gap). It shows that the spatial and temporal encoders learned in

Table 2. Performance comparisons with baselines. The reported result is the average macro F1-score across five runs by varying seeds. The labeling rate for Opportunity, RealWorld, Realdisp, and PAMAP is 10%, 1%, 10%, 10%, respectively. The best result is represented with **bold**. The runner-up's result is represented with underlined. The best result among SSL baselines is represented with *. The improvement is calculated over the best result among SSL baselines.

Approach	Opportunity	RealWorld	Realdisp	PAMAP
STMAE-Sup	52.98	63.37	88.08	<u>72.34</u>
STMAE-Rand	10.73	24.16	11.23	24.58
Masked Reconstruction [25]	50.77	61.45	81.85	68.91*
LIMU-BERT [71]	<u>53.65</u> *	62.61	83.93*	65.13
COCOA [14]	53.38	62.79	81.46	66.73
COCOA-T [†] [14]	51.34	63.57	83.09	66.51
Cosmo [51]	51.04	60.24	82.69	67.14
Cosmo-T [†] [51]	51.47	62.28	80.70	68.69
CPC [26]	52.73	<u>63.83</u> *	79.52	65.89
CPC-T [†] [26]	50.86	62.44	81.47	67.72
STMAE	56.72	65.68	<u>87.38</u>	72.54
Improvement	5.41%	2.81%	3.95%	5.00%

[†] These baselines adopt the network structure similar to STMAE's. For COCOA-T, the encoder $f_m(\cdot)$ uses the same structure of SE plus TE as in STMAE. For Cosmo-T, the feature encoder $f_{encj}(\cdot)$ uses the same structure of SE plus TE as in STMAE. For CPC-T, the encoder g_{enc} and autoregressive network g_{ar} use the same structure of SE and TE as in STMAE, respectively.

pre-training phase are practical for downstream tasks. Expectedly, STMAE-Rand has the worst performance on all datasets. The underwhelming performance of STMAE-Rand indicates the challenging nature of the activity recognition tasks we undertake, underscoring the necessity for a proficient feature extractor.

Based on the performance of SSL baselines, we observe that no single method emerges as the clear winner across all datasets. On Opportunity and Realdisp datasets, LIMU-BERT [71] achieves the best performance. On RealWorld and PAMAP datasets, CPC [26] and Masked Reconstruction [25] achieve the best performance, respectively. We note that the top-performing baselines are not those specifically designed for multi-device scenarios on all datasets. This could be because the multi-device scenarios considered in this study differ slightly from the multimodal scenarios targeted by COCOA [14] and Cosmo [51], which limits their performance. In addition, we find that masked reconstruction-based methods perform similarly to contrastive learning-based methods on average, suggesting that there is still significant potential for research on masked reconstruction-based methods.

5.2 Performance with Varying Amounts of Labeled Data

We conduct experiments to investigate the effect of labeling rate on model performance by comparing the performance of STMAE and baselines under various labeling rates. The results are presented in Fig. 2. Overall, we find that STMAE outperforms all other SSL baselines at all labeling rates for each dataset, demonstrating its adaptability to various labeling rate situations. Furthermore, compared to the supervised method STMAE-Sup, STMAE achieves similar or better performance by optimizing only a single FC layer alone in supervised learning phase. In addition, the advantage of STMAE over STMAE-Sup is more pronounced in the case of lower labeling rates, as the supervised method is prone to underfitting due to insufficient training data, while STMAE has

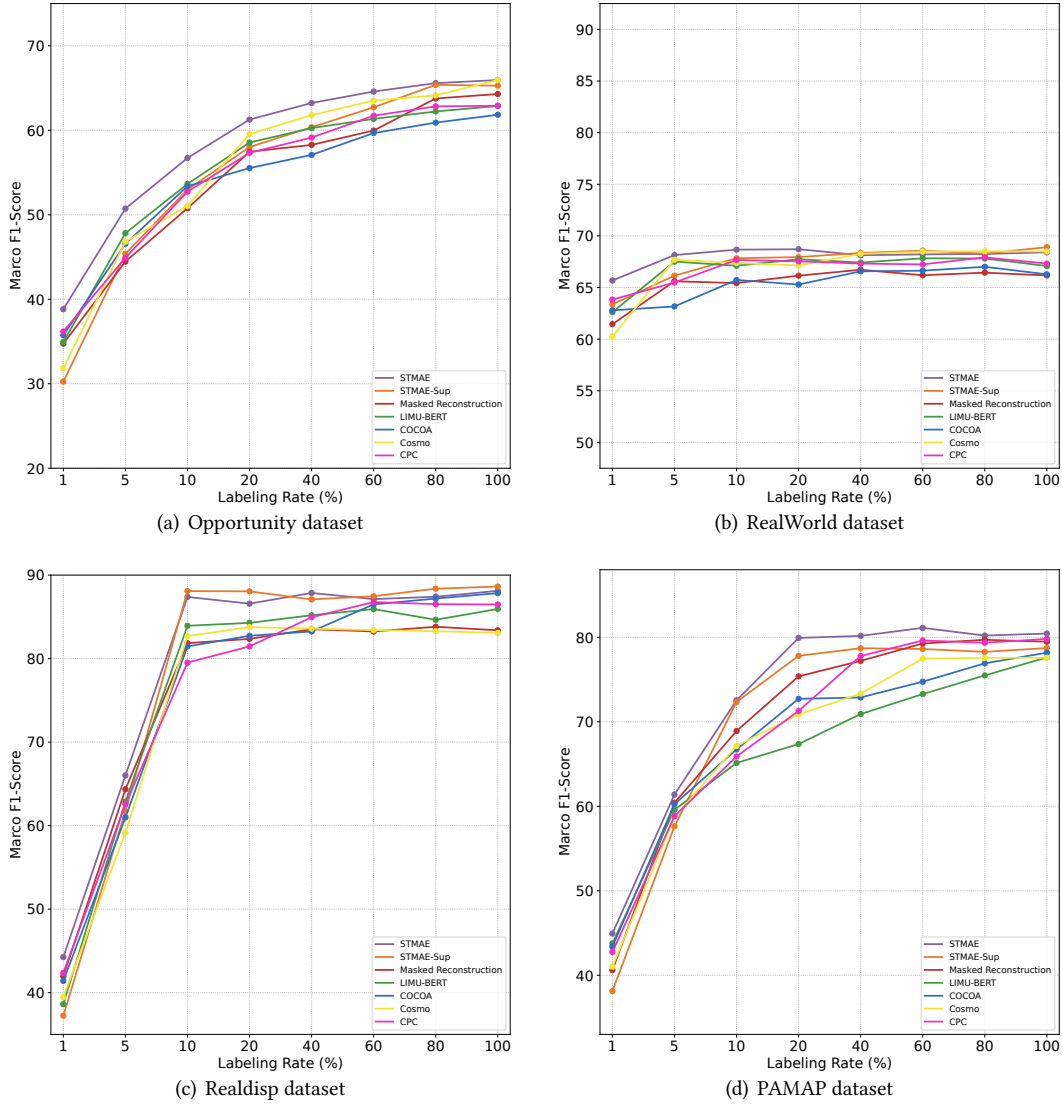


Fig. 2. Marco F1-score comparison at different labeling rates. To ensure comparability between graphs, the y-axis for each graph has been standardized to a range of 55, from the minimum to the maximum values. (Best viewed in color.)

already learned some prior knowledge of wearable data during pre-training, reducing its reliance on labeled data during supervised learning.

Our findings also reveal that macro F1-score grows rapidly with increasing labeling rate in the low labeling rate interval, where the model has a high data demand. As the labeling rate increases, the rate of increase in macro F1-score slows down. For three out of the four datasets (RealWorld, Realdisp, and PAMAP), macro F1-score eventually stabilizes at a certain labeling rate, beyond which increasing the amount of labeled data does not

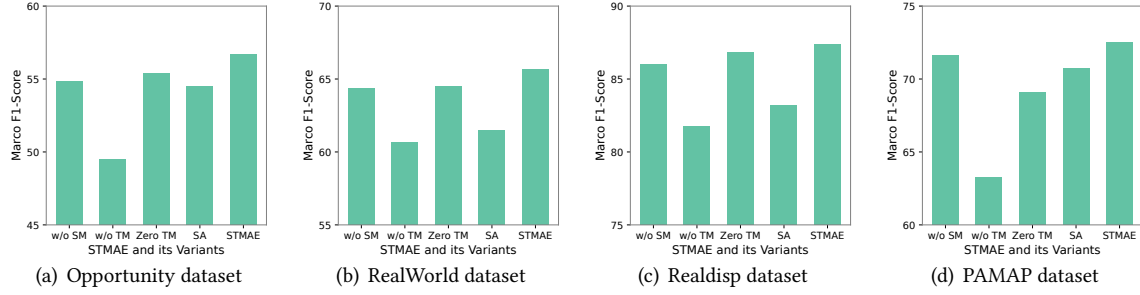


Fig. 3. Marco F1-score comparison with STMAE and its variants of masking strategies. Note that "SA" in the x-axis coordinates means symmetric autoencoder. To ensure comparability between graphs, the y-axis for each graph has been standardized to a range of 15, from the minimum to the maximum values.

result in any improvement to the macro F1-score, suggesting that the model is saturated with data and that the current amount of data are sufficient for the model to fully exploit its potential. The required amount of labeled data varies across datasets, as seen in Fig. 2. Realdisp dataset achieves stable performance at a labeling rate of 10%. However, Opportunity dataset does not seem to reach stable performance even at the maximum labeling rate of 100%. This is likely due to the differences in types of activities in each dataset. In Realdisp dataset, fitness activities are highly cyclical and have more homogeneous sub-activities, making them easier for the model to learn. For instance, "Reach heels backward" is a very fine-grained movement. In Opportunity dataset, mid-level gesture classes have more inter-class similarity and intra-class variability in their activities, resulting in confusion between similar activities, e.g., "Open Drawer 1", "Open Drawer 2", and "Open Drawer 3" that differ only in the height of the drawer.

5.3 Analysis of the Masking Strategies

We evaluate the effectiveness of the proposed spatial-temporal masking strategy by comparing its performance with several variants. Specifically, we compare with the following variants:

w/o SM: This variant removes the spatial masking and only uses the temporal masking. All elements in the spatial masked matrix M_s are set to one, and only the temporal masked data are reconstructed by RD.

w/o TM: This variant removes the temporal masking and only uses the spatial masking. All elements in the temporal masked matrix M_t are set to one, and only the spatial masked data are reconstructed by RD.

Zero TM: This variant replaces the mask token $[M]$ in temporal masking with zero value.

Symmetrical Autoencoder: Symmetric autoencoder [5, 24] is compared as a classical unsupervised learning method. The structure of encoder consists of SE and TE from STMAE for fairness and comparability. The decoder's structure is symmetrical to the encoder's structure, and no masking strategies are used. The input data are fully reconstructed by the decoder to pre-train the encoder.

The results, shown in Fig. 3, indicate that STMAE outperforms its variants on each dataset, demonstrating the effectiveness of the proposed spatial-temporal masking strategy. The performance advantage of STMAE over Zero TM indicates the effectiveness of using mask tokens in temporal masking. The performance of w/o SM is slightly inferior to that of STMAE on each dataset, while the performance of w/o TM becomes completely poor. This is because the structure of TE that corresponds to the temporal masking is Transformer encoders, which require sufficient learning to fully exploit the temporal correlations. Without the temporal masking, the temporal correlations are not competently learned, therefore the parameters of TE cannot be learned well in pre-training

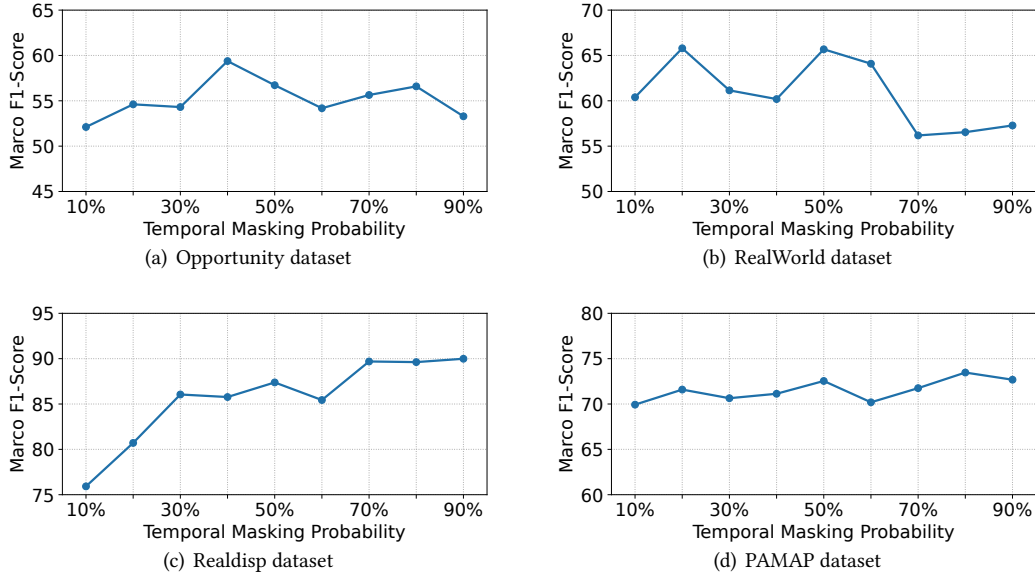


Fig. 4. Marco F1-score comparison at different temporal masking probabilities. To ensure comparability between graphs, the y-axis for each graph has been standardized to a range of 20, from the minimum to the maximum values.

phase. The spatial masking corresponds to the SE (CNN structure), which has a small number of parameters and therefore has a smaller impact on the final performance. On PAMAP dataset, we find that w/o TM performs particularly poorly, probably because its corresponding window size (300) is long, making the demand of learning temporal correlations even stronger.

For the symmetrical autoencoder, we find that it performs similarly to w/o SM on Opportunity and PAMAP datasets. It suggests that it is possible to achieve acceptable performance using only a simple autoencoder. In fact, STMAE is a further exploration on the basis of autoencoder to improve the capability of spatial-temporal correlations learning on wearable data by constructing spatial-temporal masking strategy and asymmetrical encoder-decoder structure.

5.4 Analysis of the Temporal Masking Probability

We conduct a study on how the temporal masking probability affects marco F1-score performance. As shown in Fig. 4, in general, we observe that the temporal masking probability has a significant impact on the performance of STMAE. The macro F1-score fluctuates noticeably with changes in the temporal masking probability, and this behavior depends on the dataset.

For Opportunity and RealWorld datasets, the macro F1-score initially increases as the temporal masking probability increases, reaching a sweet point, and then decreases as the temporal masking probability continues to increase. Low temporal masking probabilities provide insufficient information for the TE to learn the temporal correlations from wearable data, as the RD can easily recover the masked data. On the other hand, high temporal masking probabilities can completely destroy the temporal information in wearable data, making it difficult for the TE to be effectively trained.

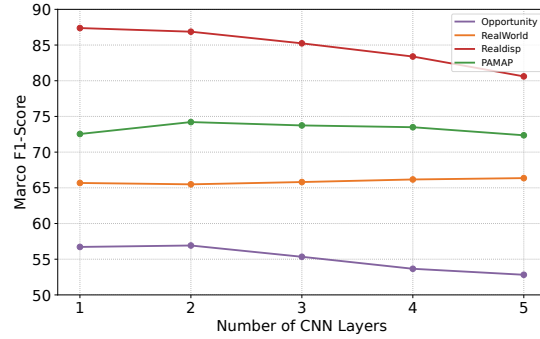


Fig. 5. Marco F1-score comparison at different settings for SE. (Best viewed in color.)

For PAMAP dataset, the macro F1-score shows a relatively gradual change with a slight upward trend. Its higher sampling frequency results in smaller variations of reading within its time window, making it easier to reconstruct. For Realdisp dataset, the macro F1-score consistently increases with increasing temporal masking probability over the entire range (from 10% to 90%). The sweet point is reached at a temporal masking probability of 90%, indicating that Realdisp dataset is not affected by the destruction of temporal properties due to high temporal masking probabilities. This can be attributed to the fine-grained nature of the fitness activities in Realdisp dataset and the strong periodicity of the corresponding wearable data, allowing for accurate reconstruction of the time series even at very high temporal masking probabilities.

The above results suggest that it is important to carefully consider the temporal masking probability when designing STMAE for specific wearable data.

5.5 Analysis of the Asymmetrical Spatial-Temporal Encoder-Decoder

5.5.1 Analysis of SE. In this study, we aim to understand the impact of the number of CNN layers on the performance of spatial information fusion among multiple devices. Our results, as shown in Fig. 5, suggest that while increasing the number of CNN layers can enhance the representation capability of the network, the impact on performance improvement is limited.

Specifically, the results indicate that for Opportunity and PAMAP datasets, the marco F1-score performance initially improves with an increasing number of CNN layers, but then gradually decreases. For Realdisp dataset, the performance decreases steadily with an increasing number of CNN layers. This decrease in performance may be due to overfitting, as the network becomes more complex with an increasing number of layers. On the other hand, RealWorld dataset exhibits slight improvement in performance with an increasing number of CNN layers, but the improvement is not significant. This could be due to the larger size of RealWorld dataset, which can still be fit well with more layers. Overall, our findings suggest that one layer of CNN can already meet the needs for spatial fusion, and that the significant improvement in performance is not necessarily achieved with an increasing number of layers.

5.5.2 Analysis of TE. In Section 5.3, we find that the temporal masking has a significant impact on the performance of STMAE. We want to further investigate whether the temporal masking still works effectively under different structures of TE. In addition, we want to determine the structure of TE that best complements the temporal masking. We compare with three alternative structures of TE with comparable network parameters: three-layer Temporal Convolutional Network (TCN) [4], three-layer Long-Short Term Memory (LSTM) [29], and three-layer FC. The input and output dimensions in each layer are all 128. The comparative experiments are conducted

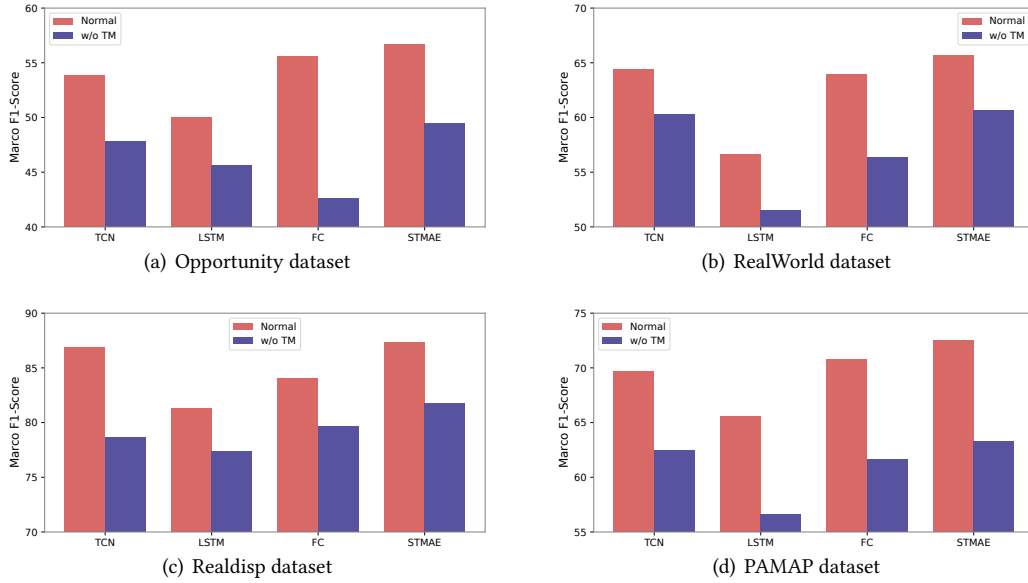


Fig. 6. Marco F1-score comparison with STMAE and its variants of TE. To ensure comparability between graphs, the y-axis for each graph has been standardized to a range of 20, from the minimum to the maximum values.

simultaneously under both the complete masking setting and w/o TM (without temporal masking) setting. The results are shown in Fig. 6.

The results demonstrate the superiority of the Transformer structure in both masking settings. It achieves the best performance among the compared structures and is capable of learning features more quickly and efficiently due to its parallel computation characteristic. On the other hand, LSTM demonstrates the worst performance, primarily due to difficulties in feature transmission within the LSTM under the masking strategy. In addition, the sequential computation method of LSTM results in slower computation speed. We also discover that the temporal masking is highly effective in various network structures, significantly improving the model's performance. Specifically, STMAE outperforms other variants and achieves the best performance. This experiment demonstrates that the combination of TE and TM plays a crucial role in the model's performance, highlighting the importance of both TE and TM themselves as well as their synergy.

5.5.3 Frozen Encoder Versus Fine-tuned Encoder. In supervised learning phase, there are generally two settings for training: frozen encoder and fine-tuned encoder. All experiments in this paper are conducted under the frozen setting, i.e., only the parameters of the classifier are optimized while the parameters of the encoders (SE and TE) are frozen. Under the fine-tuned encoder setting, both the parameters of the encoders and classifier can be optimized. To further understand the difference between these two settings, we also conduct experiments under the fine-tuned setting and compare the results with the frozen setting, as shown in Fig. 7. On Opportunity, Realdisp, and PAMAP datasets, the performance of STMAE significantly improves under the fine-tuned setting. However, on RealWorld dataset, the performance of STMAE slightly decreases under the fine-tuned setting. This may be because the labeled training data on RealWorld dataset are insufficient to fit the entire network structure of the encoders and classifier. It means that training both the encoders and classifier simultaneously during supervised learning phase is generally effective in improving model performance, but it does not guarantee

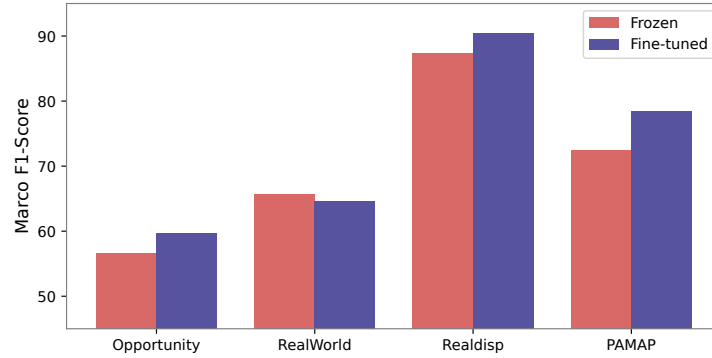


Fig. 7. Marco F1-score comparison with frozen encoder and fine-tuned encoder of STMAE. (Best viewed in color.)

complete success. Furthermore, the frozen setting alone also achieves acceptable results on all datasets. These experiments demonstrate the practicality of pre-trained encoders.

5.6 Robustness to Missing Device Scenarios

We explore the performance of STMAE under missing device scenarios, which is an inevitable problem for multi-device WHAR. It is worth noting that some experimental settings in this sub-section differ from those described in Section 4.3. During supervised learning phase, data from N_{selected} devices are randomly masked in each sample of the training set by setting the corresponding elements to zeros. N_{selected} is randomly and independently sampled from $\{0, 1, \dots, N/2\}$ for each sample. During inference phase, experiments are conducted with varying numbers of available devices. For the given number of available devices $N_{\text{available}}$, data from $N - N_{\text{available}}$ devices are randomly masked in each sample of the test set by setting the corresponding elements to zeros. The labeling rate is 100% for all datasets. The results are presented in Fig. 8. It can be seen that STMAE consistently outperforms the other baselines across all datasets for various numbers of available devices. The advantage of STMAE is particularly pronounced when only a small number of devices are available. It demonstrates the robustness of STMAE in missing device scenarios, even with only a single FC layer for supervised learning.

On Realdisp dataset, the performance degradation of STMAE with decreasing available devices is relatively minimal. However, we observe a linear decline in marco F1-score performance with decreasing available devices on the other three datasets. This suggests that while STMAE outperforms various baselines, it may still be applicable only in cases where the number of missing devices is relatively low. The specific number of applicable devices varies depending on the dataset and usage scenario. It should also be noted that no data augmentation, domain adaptation, or generative adversarial methods are used during supervised learning phase to address missing device scenarios in our study. This is because the goal of this sub-section is to demonstrate the effective and discriminative representations extracted by STMAE under missing device scenarios. It would be interesting to explore whether incorporating more advanced methods, e.g., AALH [39] and ArmTroi [45] for handling missing device scenarios in the downstream task, can lead to better performance, as the output of TE is generalized and can be utilized to diverse downstream tasks.

5.7 Discussion and Future Works

Here we provide more details on the practical implementation issues related to our method. We also present the shortcomings of the current work, as well as potential research directions for the future.

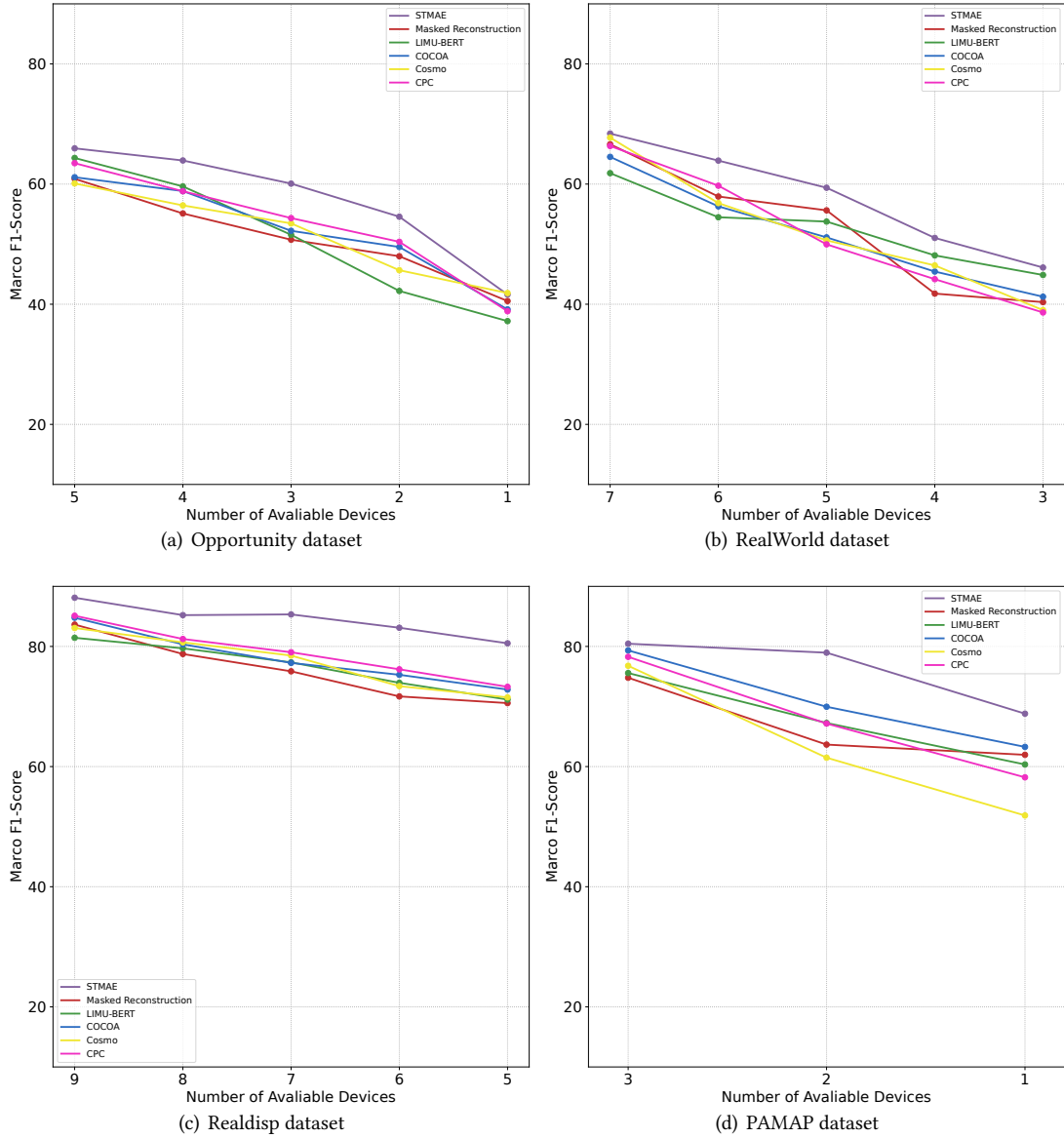


Fig. 8. Marco F1-score comparison at different number of available devices. (Best viewed in color.)

Usage Scenarios of Multi-device WHAR. Compared to single-device WHAR, multi-device WHAR requires wearing more devices, which undoubtedly affects the user's comfort and willingness to wear them. Furthermore, multi-device WHAR poses higher demands on signal stability, computational costs, and latency. In certain activity recognition tasks, single-device WHAR has already achieved acceptable performance. Therefore, the developers need to consider the scenarios and activities for which multi-device WHAR is more suitable.

Computation Overhead on Edge Devices. When deploying STMAE in practice, considerations must be given to its computation overhead on edge devices during both supervised learning and inference phases. We evaluate the computation overhead of STMAE on Opportunity dataset using a Raspberry Pi 4 Model B. In supervised learning phase, the total training time was 16.82 min, with an associated energy consumption of 47.67 KJ¹. In inference phase, each sample's inference time was 41.17 ms, with an associated energy consumption of 163.90 mJ¹. These computational costs are relatively low and meet the requirements for practical deployment. Developers may explore techniques such as pruning, quantization, distillation, and hardware acceleration to further reduce computation overhead.

Advanced Temporal Encoding. We utilize the classic Transformer encoder structure as in [66] for TE. However, recent developments in time series modeling have seen the emergence of targeted Transformer designs in positional encoding, attention module, and architecture-level [68], e.g., Autoformer [70] and Pyraformer [44]. These advanced Transformer structures may be better suited for exploring temporal features than the classic structure used in TE, and we leave this as our future work.

Pre-Training with Multi-Task. We adopt masked reconstruction [25, 28, 56, 71] for the pre-training task. However, other pre-training tasks may be introduced and combined to further enhance the ability of the feature extractor. For instance, BERT [15] uses both masked language modeling and next sentence prediction as pre-training tasks. In recent years, numerous contrastive learning-based methods have emerged in the WHAR field and have demonstrated exceptional performance [14, 26, 34, 40, 51, 54, 58, 64]. Combining masked reconstruction with contrastive learning in pre-training would be an intriguing endeavor.

Large-Scale Activity Representation Models. Despite the widespread use of self-supervised methods in the WHAR field, there are few studies on large-scale pre-training models. Currently, a separate model must be trained for each activity recognition task on each dataset. It would be meaningful to conduct pre-training on a large dataset (e.g., the UK Biobank dataset [17]) to make it applicable to multiple activity recognition tasks [76]. For instance, MotionBERT [87] is capable of performing three tasks concurrently: 3D pose estimation, skeleton-based action recognition, and mesh recovery. In our future work, we plan to develop large-scale activity representation models by utilizing wearable data.

6 CONCLUSIONS

This paper is centered on the advancement and refinement of SSL for multi-device WHAR. The primary objective of the study is to address two crucial issues present in existing SSL WHAR methods, which are dealing with missing device scenarios and constructing an effective pre-training task. We introduce STMAE, which tackles these issues by utilizing an asymmetrical encoder-decoder structure along with two-stage spatial-temporal masking strategy. STMAE leverages the spatial-temporal correlations in multi-device data to achieve superior results. Evaluation is conducted on multiple public wearable datasets, demonstrating the superiority of STMAE compared to the SOTA methods. The results indicate that even with partial device availability, STMAE can produce high macro F1-scores. This study provides a new perspective on multi-device WHAR and has the potential to inspire future research in ubiquitous computing community.

ACKNOWLEDGMENTS

The authors wish to thank the anonymous reviewers for their constructive comments on this work.

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¹We utilize the "powerstat" tool in Linux to approximate the energy consumption.

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A ACTIVITY LABELS

Here we provide the detailed activity types and their distributions in each dataset.

Opportunity: Open door 1 (5.9%), Open door 2 (6.1%), Close door 1 (5.5%), Close door 2 (5.8%), Open fridge (7.3%), Close fridge (6.7%), Open dishwasher (4.8%), Close dishwasher (4.5%), Open drawer 1 (3.6%), Close drawer 1 (2.9%), Open drawer 2 (3.4%), Close drawer 2 (2.8%), Open drawer 3 (4.1%), Close drawer 3 (3.9%), Clean table (6.7%), Drink from cup (21.9%), Toggle switch (4.3%).

RealWorld: Climbingdown (11.1%), Climbingup (13.8%), Jumping (2.2%), Lying (14.2%), Running (16.1%), Sitting (14.2%), Standing (14.0%), Walking (14.4%).

Realdisp: Walking (7.6%), Jogging (6.0%), Running (5.7%), Jump up (1.2%), Jump front & back (2.0%), Jump sideways (2.1%), Jump leg/arms open/closed (2.1%), Jump rope (1.2%), Trunk twist (arms outstretched) (4.0%), Trunk twist (elbows bended) (3.3%), Waist bends forward (3.4%), Waist rotation (3.0%), Waist bends (reach foot with opposite hand) (3.3%), Reach heels backwards (2.8%), Lateral bend (2.6%), Lateral bend arm up (2.5%), Repetitive forward stretching (1.6%), Upper trunk and lower body opposite twist (1.8%), Arms lateral elevation (2.5%), Arms frontal elevation (2.5%), Frontal hand claps (2.3%), Arms frontal crossing (2.6%), Shoulders high amplitude rotation (2.2%), Shoulders low amplitude rotation (1.4%), Arms inner rotation (1.7%), Knees (alternatively) to the breast (0.8%), Heels (alternatively) to the backside (0.9%), Knees bending (crouching) (2.9%), Knees (alternatively) bend forward (3.5%), Rotation on the knees (1.8%), Rowing (4.1%), Elliptic bike (7.2%), Cycling (7.2%).

Table 3. Hyperparameters for TE and RD.

Dataset	L_{te}	H_{te}	D_{te}	L_{rd}	H_{rd}	D_{rd}
Opportunity	5	1	64	2	4	16
RealWorld	5	2	128	1	1	64
Realdisp	5	4	128	1	4	128
PAMAP	3	4	64	1	4	32

PAMAP: Lying (10.0%), Sitting (9.6%), Standing (9.8%), Walking (12.3%), Running (5.1%), Cycling (8.5%), Nordic walking (9.7%), Ascending stairs (6.1%), Descending stairs (5.4%), Vacuum cleaning (9.1%), Ironing (12.3%), Rope jumping (2.2%).

B METHODOLOGICAL TRANSPARENCY & REPRODUCIBILITY

The source code of STMAE is available at GitHub². We conduct all the experiments in a server equipped with an Intel i9-12900k CPU, and a NVIDIA GeForce RTX 3090 GPU. The batch size in pre-training, supervised learning, and inference phase are equal to 128. In pre-training and supervised learning phase, the learning rate is set to $5e-3$. The maximum number of epochs in pre-training phase and supervised learning phase are 100 and 500, respectively, with an early stopping mechanism (training until no progress is made for five consecutive epochs). The parameter optimization of network is done by the Adam [41] optimizer. The temporal masking probability P_t is set to 50%. The number of convolution kernels D_{se} is set to 128. Zero padding with padding size (0, 2) is used in SE to preserve the size of the feature maps throughout the convolution block. The hyperparameters for TE and RD are shown in Table 3. One FC layer with the (*input_size*, *output_size*) of (D_{se} , D_{te}) and (D_{te} , D_{rd}) are utilized before the Transformer encoders of TE and RD, respectively, to adapt their corresponding hidden dimensions. The output of the Transformer encoders of RD (with the shape $\mathbb{R}^{T \times D_{rd}}$) is first passed to one FC layer with the (*input_size*, *output_size*) of (D_{rd} , $N \cdot C$) and then reshaped to get X_i^{RD} (with the shape $\mathbb{R}^{N \times T \times C}$) to fit the input shape of raw wearable data. The activity classifier consists of one FC layer with input size of D_{rd} and output size of activity class number. During supervised learning and inference phase, the output of TE is first averaged in the time dimension and then fed into the activity classifier. We obtain the official implementation of LIMU-BERT³ [71], COCOA⁴ [14], Cosmo⁵ [51], and CPC⁶ [26] from GitHub. Masked Reconstruction [25] is implemented by ourselves as no official implementation is founded.

²<https://github.com/wdkhuans/STMAE>

³<https://github.com/dapowan/LIMU-BERT-Public>

⁴<https://github.com/cruiseresearchgroup/COCOA>

⁵<https://github.com/xmouyang/Cosmo>

⁶<https://github.com/harkash/contrastive-predictive-coding-for-har>